

Structure summary

Design “fold study 1”

Generated 2026-10-06

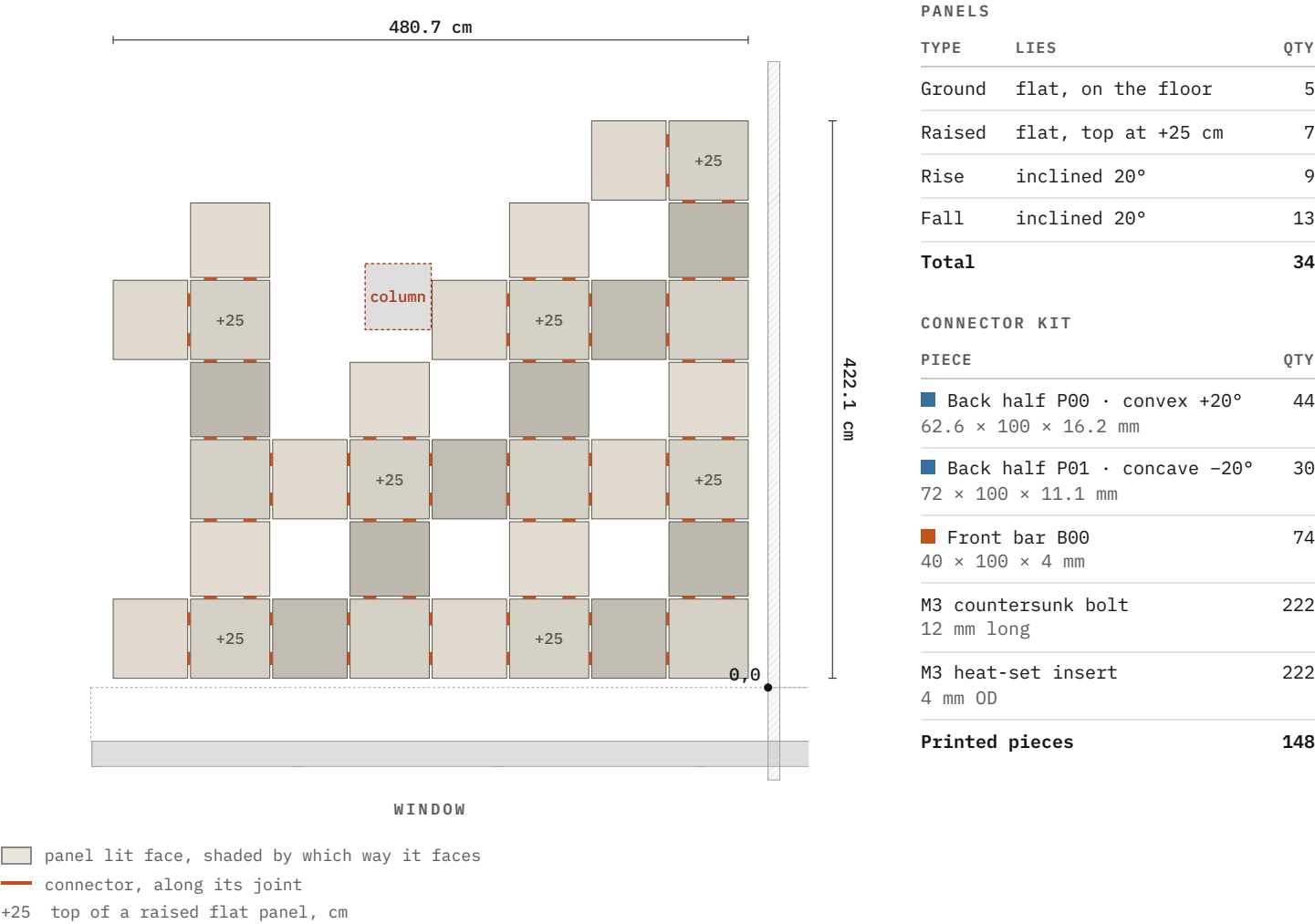
Commit 179c991 · Source design/current.json

3D model npuckett.github.io/dc-dev/grid-designer

PATTERN	FOOTPRINT	HEIGHT	PANEL GAP
trapezoid · 5 × 4 cells	480.7 × 422.1 cm	25.4 cm	22 mm

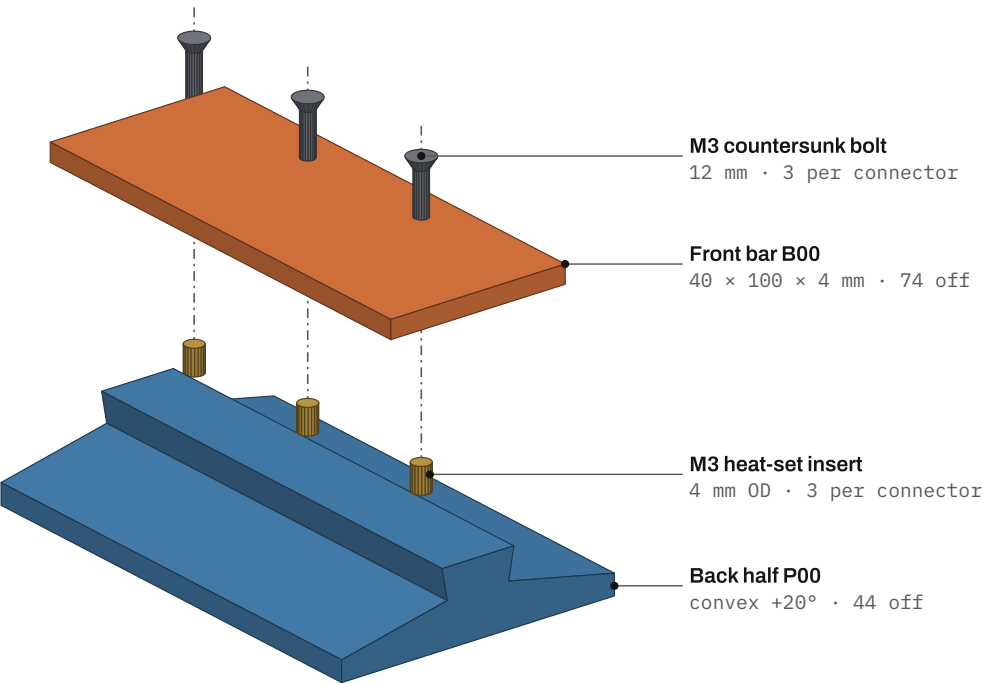
PANELS	INCLINE ANGLE	CONNECTORS
34	20°	74
22 inclined · 12 flat · 60 × 60 cm	every inclined panel · connector limit 40.6°	2 per joint · 37 joints · 148 pieces · 222 bolts

Plan from above · window at the bottom, wall on the right



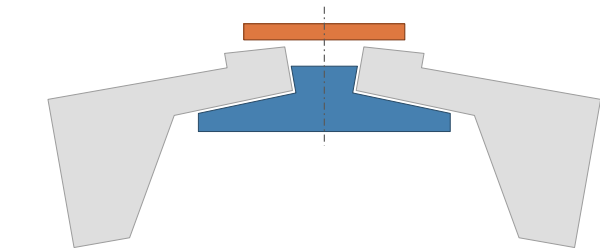
Connector two printed pieces per joint station

The back half reaches under both panel flanges. The front bar lies across both bezels. Three M3 bolts pull the two together through heat-set inserts, clamping the panel rims between them.

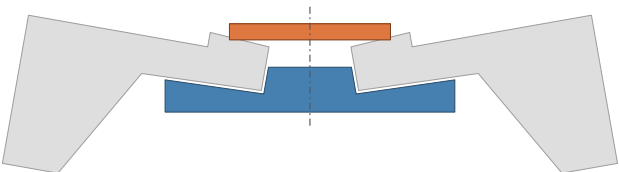


Exploded along the bolt axis. Shown: back half P00 (convex +20°) with front bar B00. Bolt positions are indicative.

ASSEMBLED, LOOKING ALONG THE JOINT



P00 · convex +20°
10 mm 44 off · built for 20 mm · joints 22 mm



P01 · concave -20°
30 off · built for 20 mm · joints 22 mm

FRONT BAR FOULS ON CONCAVE JOINTS

A flat bar clears the bezels up to 12.4° of concave fold. This design folds 20°, so the 30 concave connectors (P01) need a different bar section. The overlap shows in the P01 section above.

PIECES ROUNDED TO THE CONNECTOR BIN

Every joint here is 22 mm, but the kit rounds gaps to 5 mm steps. The back halves are built for 20 mm and the bar is 40 mm wide instead of 42. Set the editor's **span bin** to 0.20 cm or finer to build them at exactly 22 mm.

HOLES NOT MODELLED

The printable parts are solid. Bolt clearance holes and insert bores still need to be cut, and the three bolt positions along the part are not fixed yet.

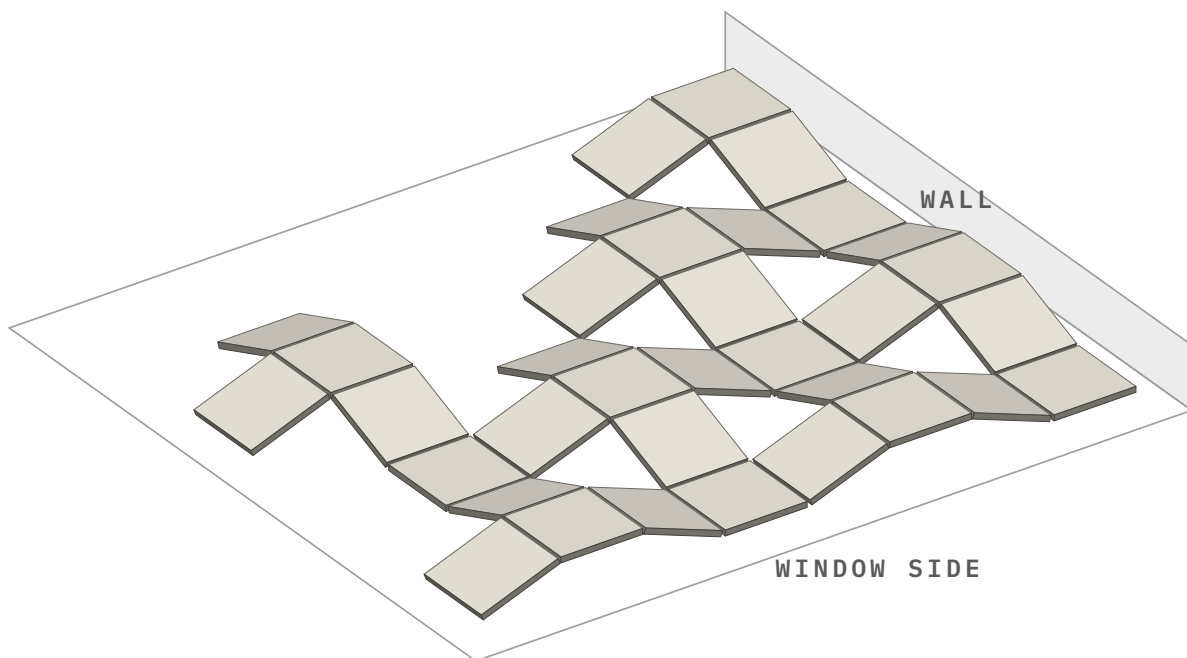
PRINTABLE PIECES

P00_back-half_convex+20deg.stl
back half · 62.6 × 100 × 16.2 mm · print 44

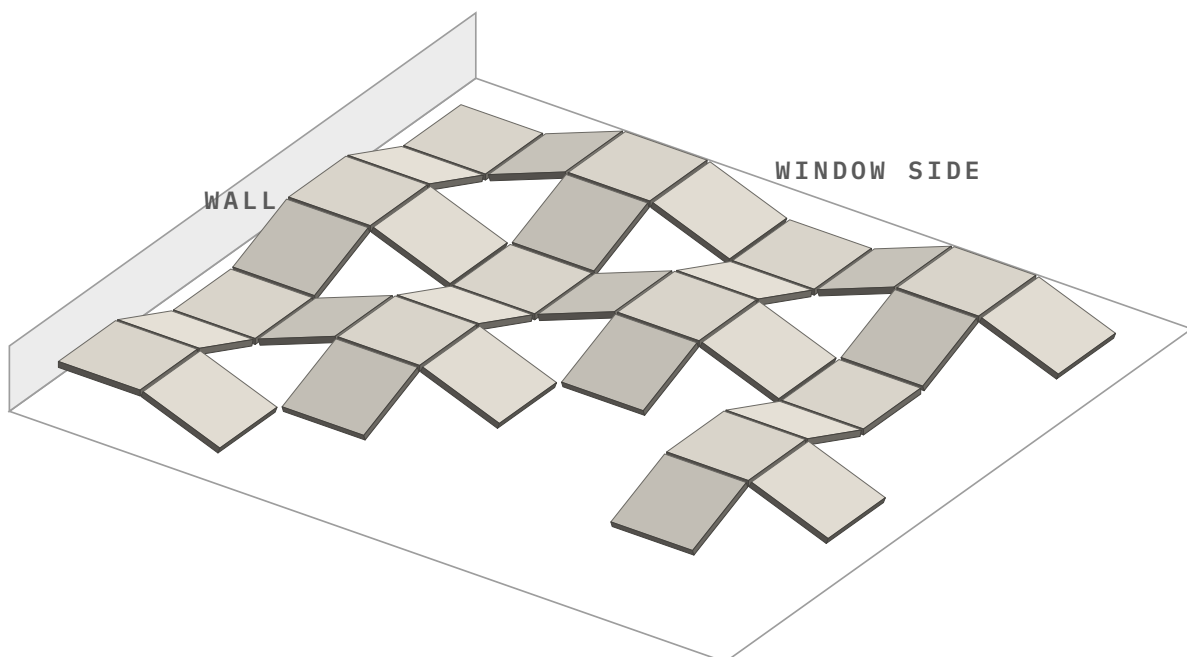
P01_back-half_concave-20deg.stl
back half · 72 × 100 × 11.1 mm · print 30

B00_front-bar_40mm.stl
front bar · 40 × 100 × 4 mm · print 74

Millimetres, laid flat for printing. In grid-designer/sheet/pieces/, or from the editor's Connector pieces button.



From the window side, wall on the right. Lit faces are pale; housings are grey.



From the room, looking back toward the window, wall on the left.
